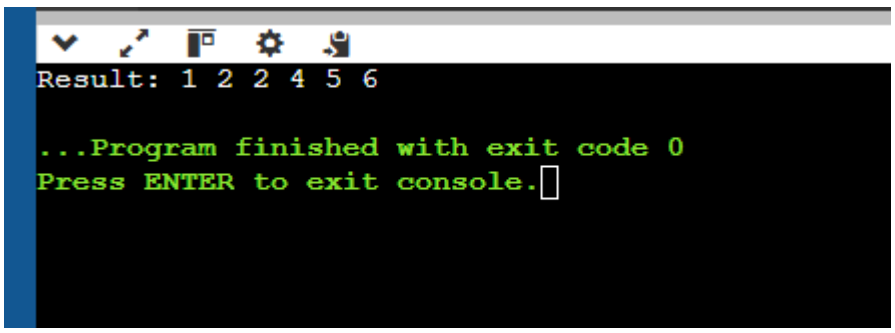
Lab3 -**2. Guided warm-up: Merge two sorted arrays**

```
#include <stdio.h>

void merge(int* nums1, int m, int* nums2, int n) {

  int i = m - 1;
  int j = n - 1;
  int k = m + n - 1;

  while (i >= 0 && j >= 0) {

    if (nums1[i] > nums2[j]) {
      nums1[k] = nums1[i];
      i--;
    }
    else {
      nums1[k] = nums2[j];
      j--;
    }

    k--;
  }

  while (j >= 0) {
    nums1[k] = nums2[j];
    j--;
    k--;
  }
```

```
}

int main() {

    int nums1[] = {2, 5, 6, 0, 0, 0};
    int nums2[] = {1, 2, 4};

    int m = 3;
    int n = 3;

    merge(nums1, m, nums2, n);

    printf("Result: ");

    for (int i = 0; i < m + n; i++) {
        printf("%d ", nums1[i]);
    }

    return 0;
}
```